

MADHAN RAJ M

PCB Design Engineer · Embedded Systems · VLSI / ECE

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TECHNICAL SKILLS

PCB Design: KiCad · Altium Designer · Proteus · Schematic Capture · 2/4-Layer Layout · RF Layout · DRC/ERC · 3D Visualization

MCU / Hardware: STM32F103C8T6 · STM32WB55CEU6 · ESP32 · Arduino · SMD & THT Soldering · Multimeter · Oscilloscope

Protocols: UART · SPI · I²C · SWD · USB · GPIO · Timers · Interrupts · ADC

VLSI / EDA: Cadence Virtuoso · Genus · Innovus · Xcelium · Xilinx Vivado · ModelSim

Programming: C · Embedded C · Verilog HDL · Python

Simulation: MATLAB · CST Studio · Proteus

PCB DESIGN PROJECTS

STM32F103C8T6 Custom Development Board | KiCad · STM32 · 2-Layer PCB

- Designed complete 2-layer PCB: schematic capture → component selection → layout → routing → DRC/ERC → 3D model
- Interfaces: USB Micro-B · AMS1117-3.3V LDO · 8 MHz crystal · SWD debug · UART · I²C · BOOT/RESET controls
- Mixed SMD + THT placement with silkscreen labeling; 4 mounting holes for mechanical fit
- Verified with ERC/DRC checks; delivered: schematic, PCB layout, 3D board visualization

STM32WB55CEU6 Wireless & RF PCB Design | KiCad · RF Layout · 4-Layer PCB

- Designed 4-layer wireless PCB with dedicated power/ground planes for signal integrity
- RF section: u.FL connector · RF matching network · ground/copper-plane implementation
- Dual crystal: 32 MHz HSE + 32.768 kHz LSE · USB-C (5V→3.3V, MIC5365) · Tag-Connect SWD
- Fiducial markers for SMT alignment; deliverables: schematic, layer views, 3D model

5V Buck Converter PCB | Altium Designer · Power Electronics

- Designed DC-DC buck converter schematic and PCB layout in Altium Designer
- Assembled with SMD + THT soldering; validated output with multimeter and oscilloscope

EMBEDDED SYSTEMS PROJECTS

Cardiac Health Monitoring System | ESP32 · ThingSpeak · Embedded C

- Real-time heart rate monitoring with ESP32 + biomedical sensors; wireless data to ThingSpeak IoT platform
- Implemented ADC sampling, interrupt-based event detection, and Wi-Fi transmission

Bluetooth Controlled Robotic Car | Arduino · L298N · Bluetooth

- Built wireless robotic vehicle with Arduino + L298N motor driver; Bluetooth smartphone control
- Interrupt-driven firmware for real-time directional control: forward, reverse, left, right

VLSI & RESEARCH PROJECTS

Energy-Efficient Approximate Full Adders | Verilog HDL · MATLAB

- Implemented RCPFAIV/V and ERPFA-III/IV designs; 8-bit hybrid adder (approximate + exact stages)
- Achieved 18–26% dynamic energy reduction · 35% static power reduction
- Validated via MATLAB Gaussian image filtering — PSNR: 49–53 dB

Metasurface-Based Multilayer UWB Antenna | CST Microwave Studio

- Multilayer UWB antenna on Rogers substrate · 3.1–10.6 GHz · return loss < −10 dB
- Via-wall structures for wideband operation; targeted 5G/6G wireless applications

EXPERIENCE

Embedded Systems & VLSI Intern · QMOS Technology, Chennai

Dec 2025 – Feb 2026

- Firmware: ESP32/Arduino sensor integration via GPIO, ADC, timers, interrupts; 5+ sensors over UART, SPI, I²C
- PCB simulation in Proteus and Altium; hardware debugging with multimeter and oscilloscope
- RTL workflows: approximate full adder concepts; MATLAB image processing for hardware validation

VLSI Intern · NSIC2025

- Designed and verified 8+ digital modules (MUX, adders, counters) in Verilog HDL using ModelSim — ↓40% debug iterations
- Synthesized and deployed RTL designs on Xilinx FPGA boards; verified timing and functionality

EDUCATION

B.E. Electronics & Communication Engineering

Meenakshi Sundararajan Engineering College, Chennai **CGPA: 8.49 / 10**

Nov 2022 – Jun 2026

Class XII — State Board

Jaihind Matriculation School **86.5%**

2022

Class X — State Board

Jaihind Matriculation School **98.8%**

2020

CERTIFICATIONS

- NPTEL: CMOS Digital VLSI Design · VLSI Design Flow (RTL to GDS) · Microprocessor & Microcontroller
- AMD–Xilinx Webinar: Vitis Platform, Zynq MPSoC Architecture
- MATLAB OnRamp: Signal Processing · Image Processing

ACHIEVEMENTS & ACTIVITIES

- Top 20 / 3,500 teams — Hackz2024, CEG Anna University
- Placement Coordinator · Symposium Coordinator